

TASK 2 - OPTIMAL SEED SELECTION

From Terminal:

Seed 0:

```
Min p-value: 0.2075
Age_train_vs_validation: 0.2075
Age_train_vs_test: 0.3009
Age_validation_vs_test: 0.7659
Annual Income (k$)_train_vs_validation: 0.8570
Annual Income (k$)_train_vs_test: 0.8570
Annual Income (k$)_validation_vs_test: 0.5786
Spending Score (1-100)_train_vs_validation: 0.6389
Spending Score (1-100)_train_vs_test: 0.9546
Spending Score (1-100)_validation_vs_test: 0.7659
*** FOUND IDEAL SEED 0 - all p-values > 0.05 ***
```

--- covariate shift check results ---

seed: 0

p-values for ks tests:

```
Age_train_vs_validation: 0.2075
Age_train_vs_test: 0.3009
Age_validation_vs_test: 0.7659
Annual Income (k$)_train_vs_validation: 0.8570
Annual Income (k$)_train_vs_test: 0.8570
Annual Income (k$)_validation_vs_test: 0.5786
Spending Score (1-100)_train_vs_validation: 0.6389
Spending Score (1-100)_train_vs_test: 0.9546
Spending Score (1-100)_validation_vs_test: 0.7659
```

KS test p-values for Selected Seed: 0:

- I. Age: 0.2075–0.7659
- II. Annual Income (k\$): 0.5786–0.8570
- III. Spending Score (1-100): 0.6389–0.9546

Explanation:

I iterated through random seeds and found that Seed 0 yielded all p-values > 0.05, indicating no statistically significant covariate shift among the splits. This means the feature distributions are consistent across the training, validation, and test sets, ensuring unbiased clustering analysis.

TASK 3 - EDA RESULTS:

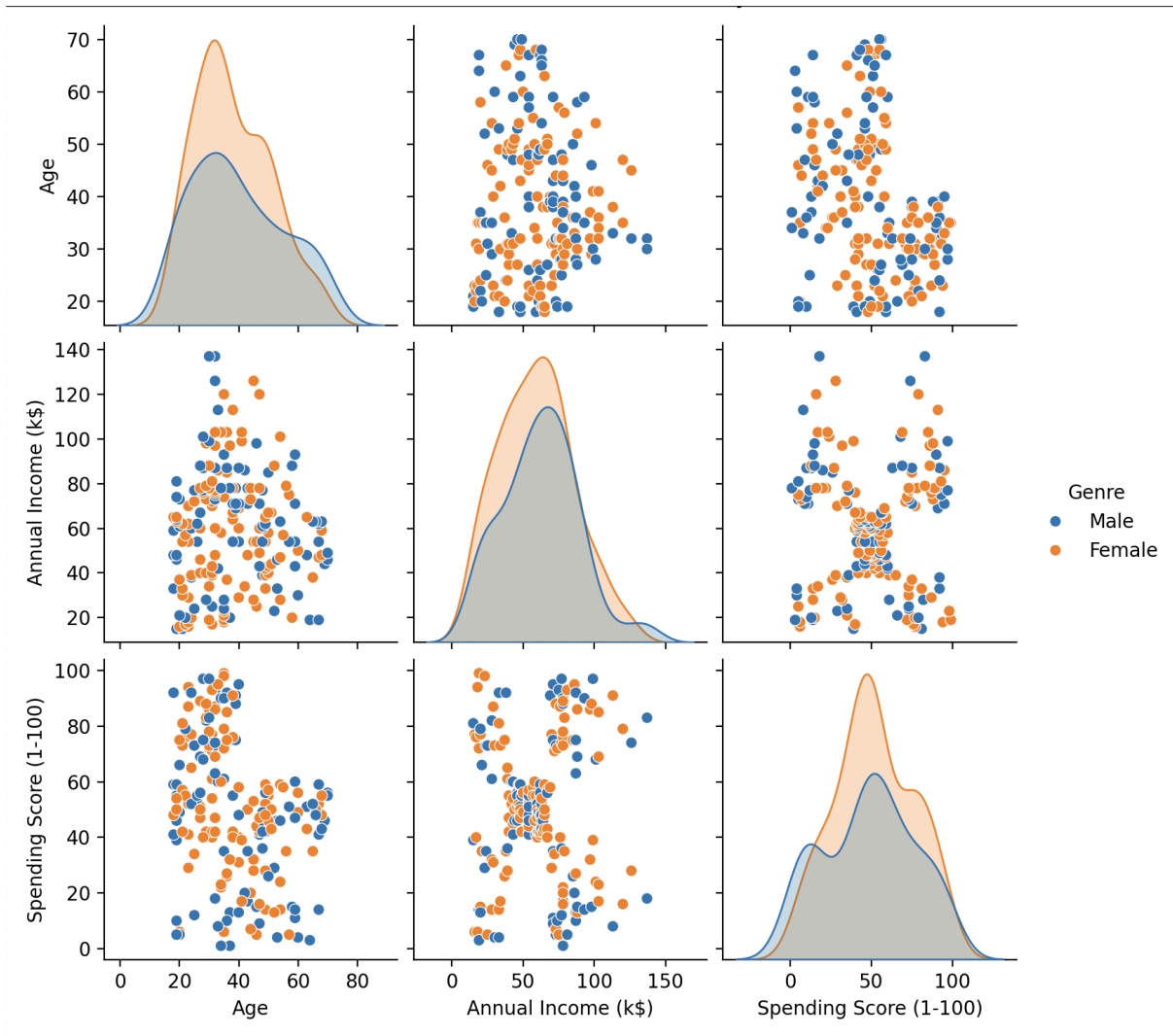
1. Summary Statistics

From Terminal:

--- SUMMARY STATISTICS FOR NUMERICAL FEATURES ---

	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000
mean	38.850000	60.560000	50.200000
std	13.969007	26.264721	25.823522
min	18.000000	15.000000	1.000000
25%	28.750000	41.500000	34.750000
50%	36.000000	61.500000	50.000000
75%	49.000000	78.000000	73.000000
max	70.000000	137.000000	99.000000

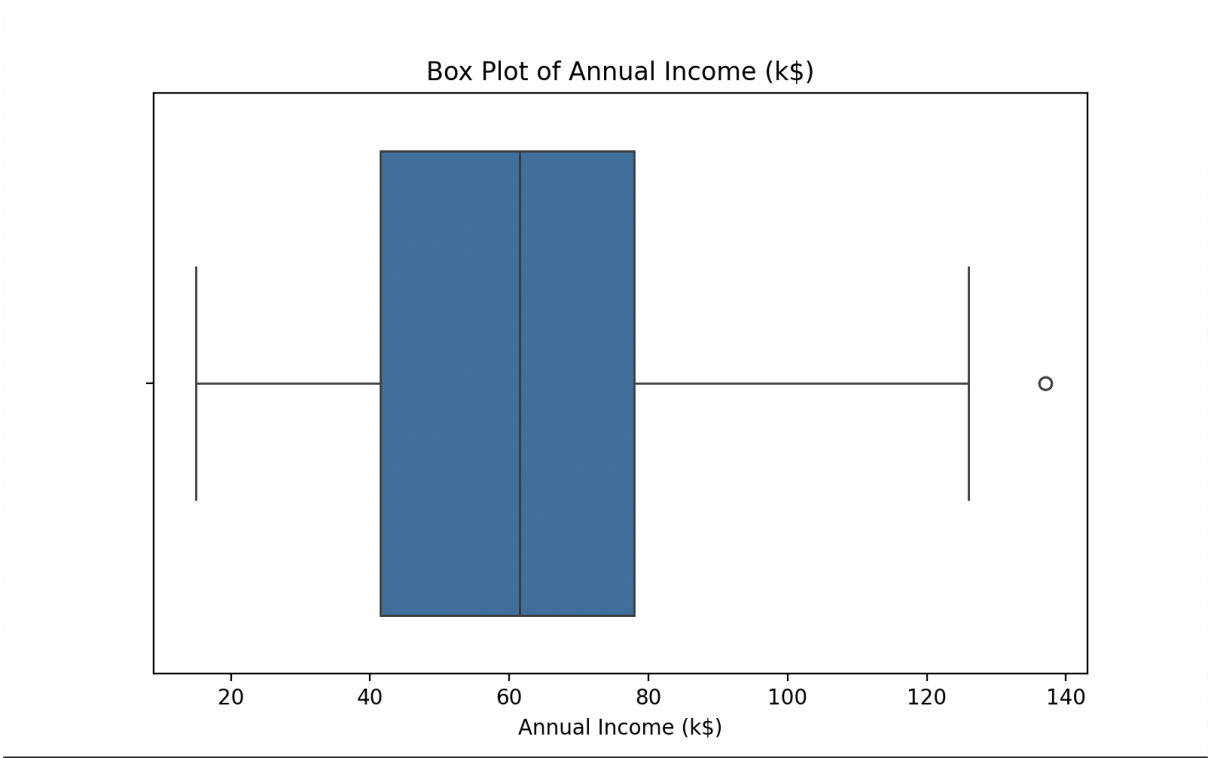
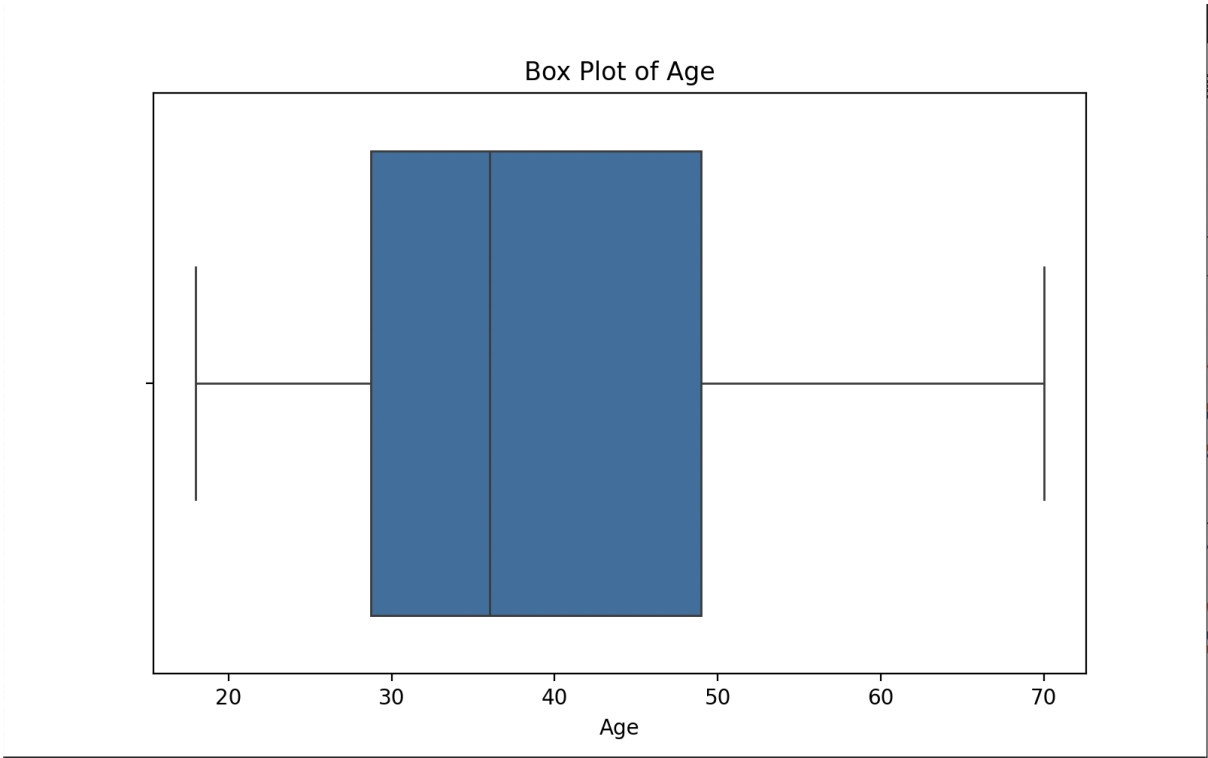
2. Visualizations

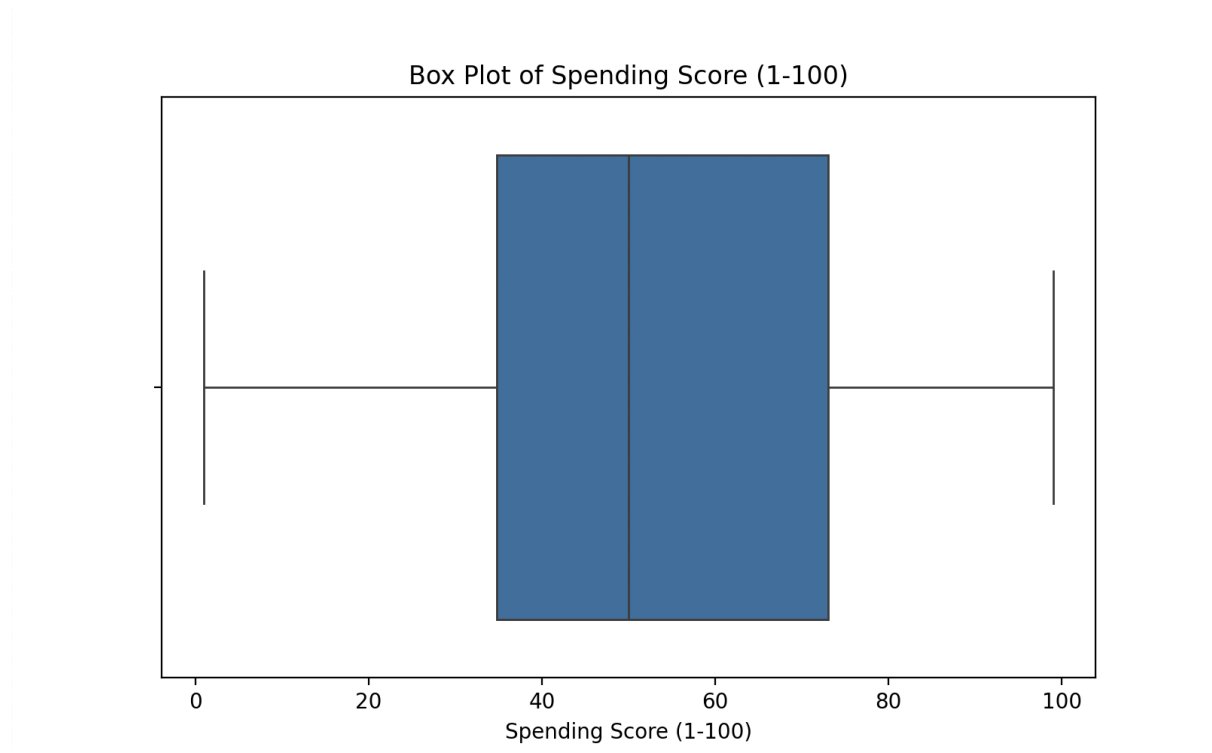


The visualization shows the relationships between Age, Annual Income, and Spending Score, where data points are colored by Gender (blue for Male, orange for Female). The diagonal plots show that the distributions of all three features differ slightly by gender. The plots show some interesting patterns. For example, Spending Score varies widely among customers with similar income levels, and younger individuals show a wider range of spending behavior compared to older ones. These patterns display variability in the data.

3. Outlier Detection

Box Plot Outlier Detection:





Z-Score Outlier Detection:

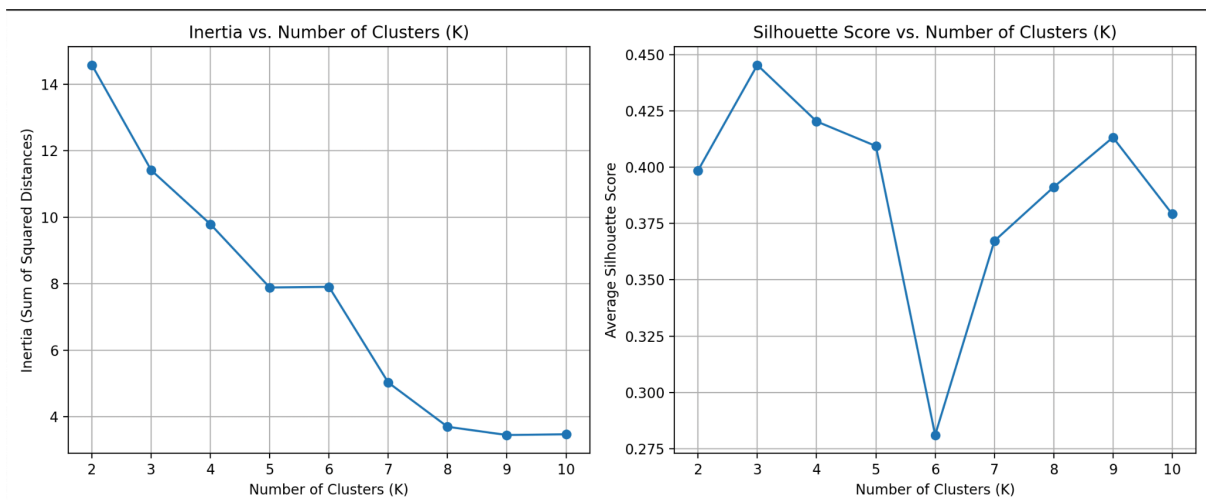
From Terminal:

```
--- outlier detection using z-scores (|z| > 3) ---  
\potential outliers in 'Age':  
no outliers in 'Age' line71  
\potential outliers in 'Annual Income (k$)':  
no outliers in 'Annual Income (k$)' line71  
\potential outliers in 'Spending Score (1-100)':  
no outliers in 'Spending Score (1-100)' line71  
  
no outliers in any feature line 74
```

Explanation:

Outlier detection was performed using both the z-score method and box plots. The z-score method, which flags data points beyond three standard deviations from the mean, did not identify any outliers in the dataset. In contrast, the box plot method, which is based on quartile thresholds, identified one outlier in the Annual Income variable. Despite this, the outlier was retained in the dataset because it likely represents a high-earning customer and z-score did not identify this outlier.

TASK 5 - HYPERPARAMETER TUNING



From Terminal:

--- Hyperparameter Tuning for K (K from 2 to 10) ---

```
training k-means K=2
converged at iteration 3 distance metric: euclidean
inertia for K=2: 14.5802
Silhouette Score for K=2 on validation set: 0.3986
training k-means K=3
converged at iteration 3 distance metric: euclidean
inertia for K=3: 11.4162
Silhouette Score for K=3 on validation set: 0.4454
training k-means K=4
converged at iteration 11 distance metric: euclidean
inertia for K=4: 9.7902
Silhouette Score for K=4 on validation set: 0.4203
training k-means K=5
converged at iteration 7 distance metric: euclidean
inertia for K=5: 7.8876
Silhouette Score for K=5 on validation set: 0.4094
training k-means K=6
converged at iteration 5 distance metric: euclidean
inertia for K=6: 7.9063
Silhouette Score for K=6 on validation set: 0.2811
training k-means K=7
converged at iteration 8 distance metric: euclidean
inertia for K=7: 5.0285
Silhouette Score for K=7 on validation set: 0.3673
training k-means K=8
converged at iteration 11 distance metric: euclidean
inertia for K=8: 3.6984
Silhouette Score for K=8 on validation set: 0.3912
training k-means K=9
converged at iteration 7 distance metric: euclidean
```

inertia for K=9: 3.4508
Silhouette Score for K=9 on validation set: 0.4133
training k-means K=10
converged at iteration 5 distance metric: euclidean
inertia for K=10: 3.4724
Silhouette Score for K=10 on validation set: 0.3792

PLOTTING METRICS

TASK 5 COMPLETE

K values tested: [2, 3, 4, 5, 6, 7, 8, 9, 10]

Summary of Results

Inertia Values:

- K=2: 14.58
- K=3: 11.42
- K=4: 9.79
- K=5: 7.89
- K=6: 7.91
- K=7: 5.03
- K=8: 3.70
- K=9: 3.45
- K=10: 3.47

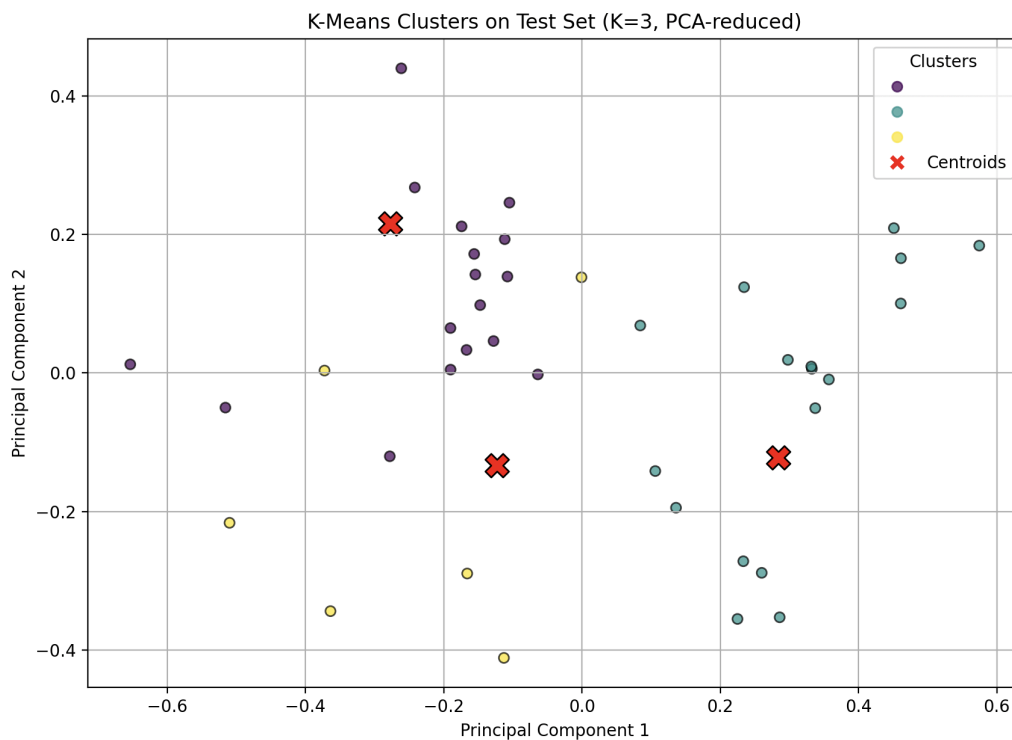
Silhouette Scores:

- K=2: 0.3986
- K=3: 0.4454
- K=4: 0.4203
- K=5: 0.4094
- K=6: 0.2811
- K=7: 0.3673
- K=8: 0.3912
- K=9: 0.4133
- K=10: 0.3792

Explanation:

After looking at both the inertia and silhouette scores, I chose $K=3$ as the best number of clusters. Inertia shows that adding more clusters after 3 doesn't help much, which is called the "elbow point." This means that with $K=3$, the model already captures the main patterns in the data without making things too complex. The silhouette score is also highest at $K=3$, which means the clusters are clearly separated and make sense. Even though there are other K s with okay scores, none are better than $K=3$. So, picking $K=3$ gives a good balance between accuracy and keeping the model simple.

TASK 6 - EVALUATION ON TEST SET



From Terminal:

EVALUATION

Inertia on Test Set: 3.3070946960726886

Silhouette Score (Scratch) on Test Set: 0.3509397606956203

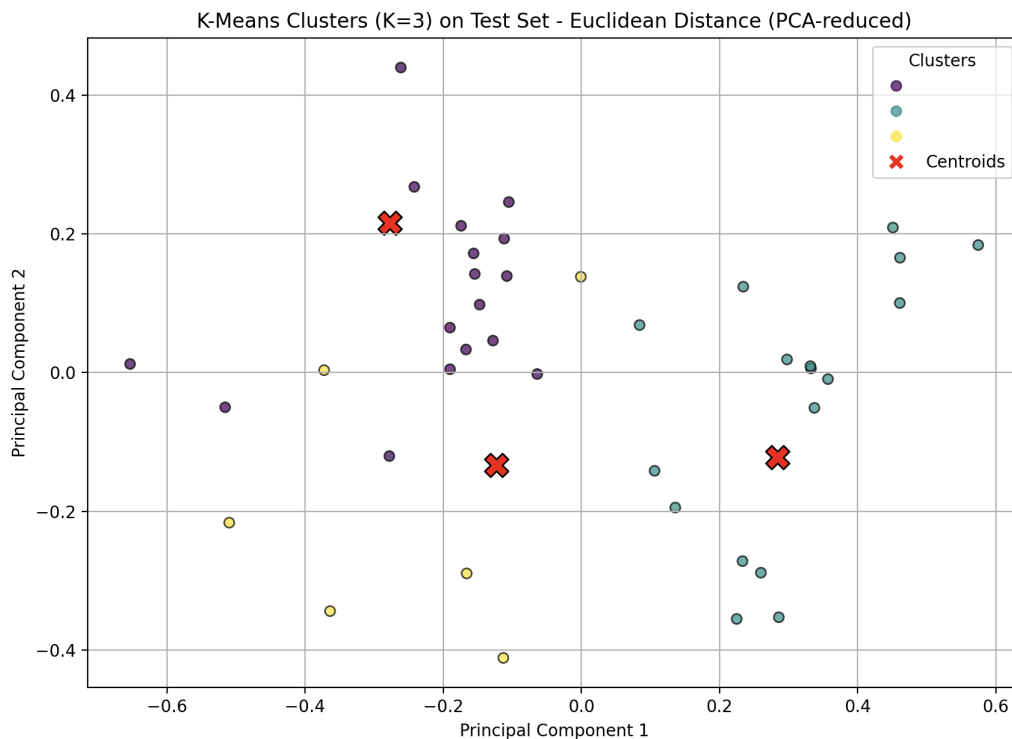
Silhouette Score (Scikit-learn) on Test Set: 0.3509397606956203

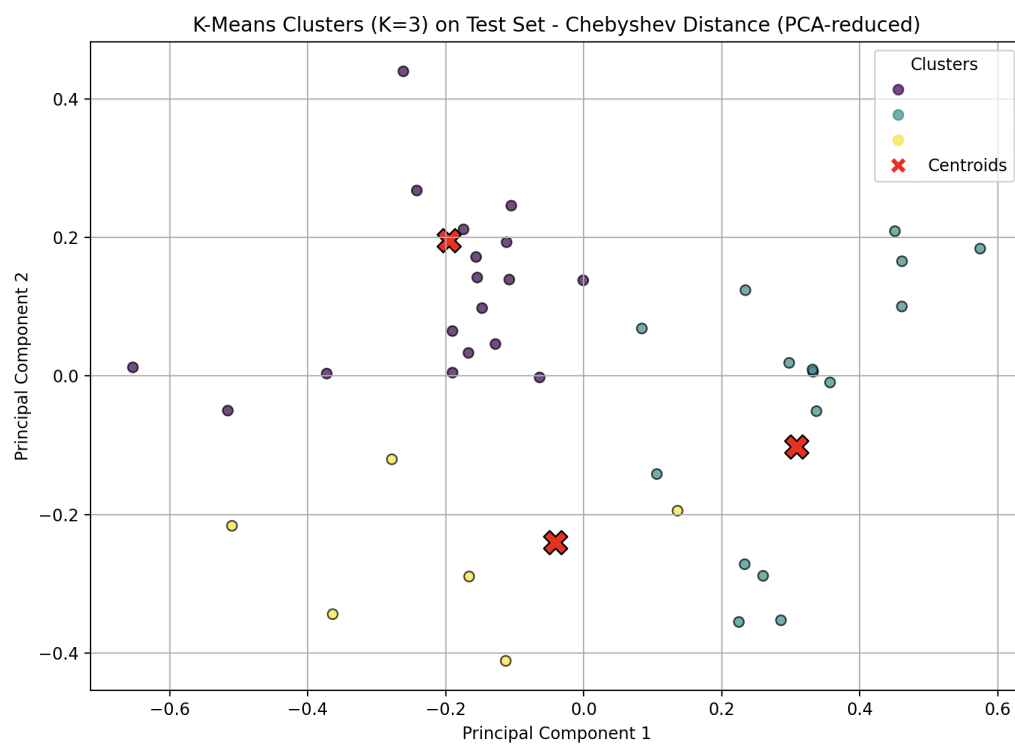
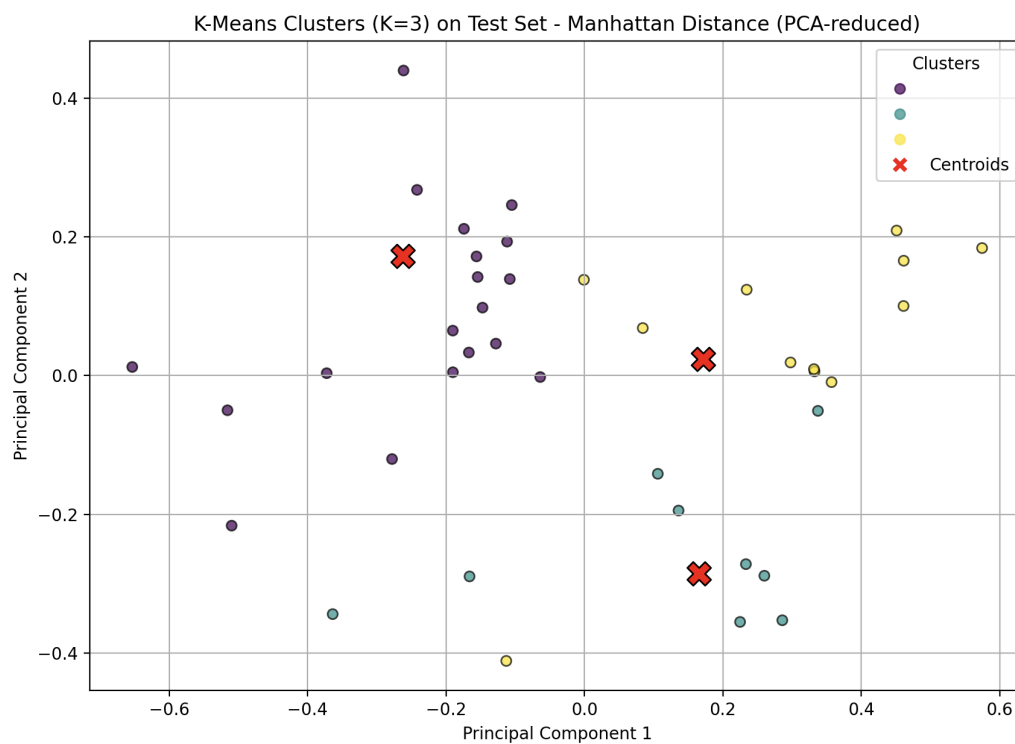
Explanation:

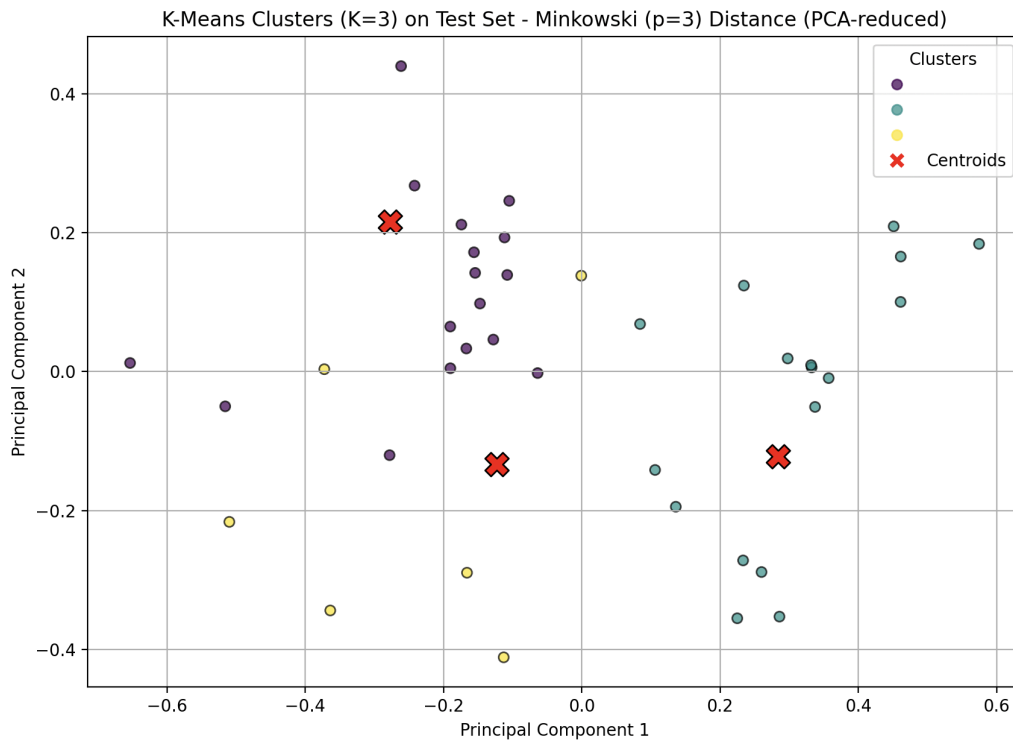
We tested our K-Means model on the test set using $K=3$, which we chose as the best number of clusters earlier. The inertia was 3.31, showing that points are close to their cluster centers. The silhouette score was 0.351, and it matched the score from scikit-learn, so our implementation is working correctly.

We also used PCA to reduce the data to 2D and plotted the clusters. The plot shows three clear groups, with most points close to their centroids, although one cluster looks a bit more spread out. Overall, the clustering looks good using Euclidean distance, and $K=3$ performs well on the test set.

BONUS TASK - ALTERNATIVE DISTANCE METRICS







From Terminal:

EUCLIDEAN:

EVALUATION WITH Euclidean

TRAINING with Euclidean

converged at iteration 3 distance metric: euclidean

training complete

EVALUATION ON TEST SET Euclidean

Inertia for Euclidean: 3.3071

Silhouette Score (Scratch) for Euclidean: 0.3509

VISUALIZING CLUSTERS FOR Euclidean

MANHATTAN

EVALUATION WITH Manhattan

TRAINING with Manhattan

converged at iteration 13 distance metric: manhattan

training complete

EVALUATION ON TEST SET Manhattan

Inertia for Manhattan: 3.4647

Silhouette Score (Scratch) for Manhattan: 0.3561

VISUALIZING CLUSTERS FOR Manhattan

CHEBYSHEV

EVALUATION WITH Chebyshev

TRAINING with Chebyshev

converged at iteration 16 distance metric: chebyshev

training complete

EVALUATION ON TEST SET Chebyshev

Inertia for Chebyshev: 3.3733

Silhouette Score (Scratch) for Chebyshev: 0.3249

VISUALIZING CLUSTERS FOR Chebyshev

MINKOWSKI

EVALUATION WITH Minkowski (p=3)

TRAINING with Minkowski (p=3)

converged at iteration 3 distance metric: minkowski

training complete

EVALUATION ON TEST SET Minkowski (p=3)

Inertia for Minkowski (p=3): 3.3071

Silhouette Score (Scratch) for Minkowski (p=3): 0.3436

VISUALIZING CLUSTERS FOR Minkowski (p=3)

--- SUMMARY OF BONUS TASK RESULTS (K={OPTIMAL_K}) ---

	metric	...	silhouette
0	Euclidean	...	0.350940
1	Manhattan	...	0.356082
2	Chebyshev	...	0.324890
3	Minkowski (p=3)	...	0.343649

[4 rows x 3 columns]

Explanation:

In this task, I implemented K-Means clustering using three alternative distance metrics—Manhattan, Chebyshev, and Minkowski (p=3)—in addition to the standard Euclidean distance. I modified the K-Means algorithm to accept a distance metric parameter. Manhattan distance gave the best silhouette score, meaning it formed slightly better clusters, but it took more steps to finish. Euclidean and Minkowski (p=3) were almost the same in how well they grouped the data and finished quickly in just 3 steps, so they were fast and gave good results. Chebyshev had the lowest score and took the most steps, which means its clusters were not as clear. Overall, Manhattan made the best clusters, but Euclidean and Minkowski were faster and still worked well.